

LONG Question

Q. What are the responsibilities of database administration?

Database administrator:

⇒ A database administrator is responsible for maintaining, securing, and operating databases and also ensures that data is correctly stored and retrieved.

⇒ There are several types of database administrators, each with specific duties and responsibilities.

⇒ The most common types of DBAs include system administrators, database architects, database analysts, data modelers, application DBAs, task oriented DBAs, performance analysts, data warehouse administrators, and Cloud DBAs.

Responsibilities of DBAs

1. Database Design and Development:

- Design and implement database structures, schema, and architecture.
- Ensure data normalization, scalability, and performance.
- Develop and maintain data models, data dictionaries, and data flow diagrams.

2. Database installation and configuration:

- Install and Configure database management system like Oracle, MySQL etc.
- Setup database environments, including development, testing, and production.
- Configure database settings, parameters and options for optimal performance.

3. Data Management & Security:

- Manage data storage, retrieval and backup.

- Ensure data consistency, integrity, and security.
- Implement data encryption, access controls and authentication mechanisms.

4. Performance Tuning and optimization:

- Monitor and optimize database performance, queries and indexing.
- Troubleshoot and resolve performance issues.
- Use database tuning tools and techniques to improve the efficiency.

5. Backup and Recovery:

- Develop and implement backup and recovery strategies.

- Ensure data availability and data loss.
- Test and validate backup and recovery processes.

6. Database Maintenance and Upgrades:

- Perform routine database maintenance tasks, such as updating statistics, running DBCC checks, and rebuilding indexes.
- Plan and execute database upgrades, migrations, and patches.
- Ensure compatibility and resolve issues after upgrades.

7. Query Optimization and Troubleshooting:

- Analyze and optimize SQL queries for better performance.
- Troubleshoot and resolve database issues, errors and failures.

- Use database debugging tools and techniques to identify and fix problems.

8. Documentation and Reporting:

- Maintain database documentation, including schema, data models, and data dictionaries.
- Generate reports and metrics on database performance, usage and trends.
- Provide insights and recommendations for database improvements.

9. Collaboration and Support:

- Works with developers, analysts and other teams to ensure database alignment with business requirements.
- Provide database support and training to users and teams.
- Collaborate with IT teams for infrastructure, network and security support.

What is schema?

- The skeleton of the database is created by the attributes and this skeleton is named schema.
- Schema mentions the logical constraints like table, primary key, etc.
- The schema does not represent the data type of the attributes.
- Database schema is logical representation of data that shows how the data in a database should be stored logically.
- Define how the data is organized and the relationship b/w the tables.

Types of database schema:

Logical Physical view

What is view?

⇒ View is a virtual table that does not physically exist. It is stored in Oracle data dictionary and does not store any data. Can be executed when called.

Q. What is tablespace?

• Oracle stores data logically in tablespaces and physically in datafiles.

• An Oracle database consists of one or more logical storage units called tablespaces.

• The database's data is collectively stored in the database's tablespaces.

• Each tablespace consists of one or more files called data files.

• A database is divided into one or more logical storage units called tablespaces.

• A DBA can use tablespace, add datafiles to tablespaces, set and alter default segment storage settings for segments created in a tablespace.

• Tablespaces are divided into logical units storage called segments.



Q. What is data block?

- A data block is the smallest unit of logical storage for a database object.

- As the object grows they take chunks of additional storage that are composed of contiguous data blocks.

Define segment.

- A segment is a set of extent.

- Always stored in single tablespace.

- Each segment is divided into extents and extents are divided into data blocks.

Q. Define extent.

- Extent consist of one or more database blocks.

